Step3:

Formula: $P(r,m,t,a) = P(r) \times P(m|r) \times P(t|r,m) \times P(a|t)$

$= 0.7 \times 0.6 \times 0.9 \times 0.9$

$P = 0.3402$

Step4 (Q#8)

$0.1 \times 0.1 \times 0.6 \times 0.4$

$= 0.0024$

Step6:

No rain: $0.7(0.4 \times 0.2 + 0.6 \times 0.1) = 0.098$

light rain: $0.2(0.2 \times 0.4 + 0.8 \times 0.3) = 0.064$

heavy rain: $0.1(0.1 \times 0.6 + 0.9 \times 0.5) = 0.051$

Q11: $0.098 + 0.064 + 0.051 = 0.213$

Q12: $0.098 \div 0.213 = 0.4601$

Short answer:

13) because the code performs the same exact calculations as the hand method, just faster.

14) because appointment depends only on Train, so when we know the train's delayed, we can directly get the probability from the table.

16) Because missing the appointment suggests that the train was delayed, and delays are more likely when there is rain.

18) Because with heavy rain, the probability of a delay is already high, whether maintenance happens or not. So, knowing the train was delayed does not tell us much about maintenance.